Lab 1 - TrashTag Product Description

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1 Introduction

With more than $40 million dollars of taxpayer money being spent to clean up litter in Texas each year (Garza, 2016), it is crucial that this money be used effectively to not unnecessarily burden taxpayers. However, this money is not being used efficiently as it takes many volunteers hundreds of hours to be able to locate and clean up litter. For example, more than 625 volunteers were needed to clean up 8,000 lbs of litter in 2023 (Fawaz, 2023). While their effort is commendable, this raises the question of how much more waste that volunteers would have been able to collect if they did not have to spend so much time searching for where the waste was dumped.

Illegal dumping also presents significant environmental challenges. Oversized items such as mattresses, tires, and appliances are among the most difficult and dangerous to clean up. Their removal often requires many volunteers and sometimes specialized equipment that local governments may not have access to without significant advance planning. However, when this litter is not seen before the cleanup is organized, it can waste volunteers' time and taxpayers' money.

Beyond the logistical challenges, litter poses direct environmental and health hazards to tourists and residents. Beaches are a common area where citizens get injuries as just under one quarter of beach goers get injured by litter each year (Campbell et al., 2016). These monetary and environmental problems can be significantly reduced with the right solution. TrashTag aims to be the solution that local governments, volunteers, and non-profit organizations are looking for.

TrashTag is a browser application that helps local governments and volunteers tag and clean up trash by organizing clean up groups. By connecting local governments, non-profit organizations, and volunteers through a single platform, TrashTag aims to transform litter cleanup from a costly, disorganized effort into an efficient, community-driven initiative.

2 Product Description

2.1 Key Products and Features

TrashTag is a browser application designed to improve the waste cleanup process by allowing users to photograph and report litter and organize and conduct cleanups. Live reporting will be available for users to get real-time litter reports and users will be alerted to nearby litter. Images of litter will be provided to users as well as verification of whether the litter is still there or not. A ranking system will be used as gamification for the cleanup process. Users will have access to a leaderboard to see where they rank against other users in their area or even worldwide.

TrashTag allows users to take pictures of trash, pin its location on a map, upload it to the website, and organize clean up groups to help dispose of illegally dumped litter. Local governments will spend less money searching for litter when people in the community can tag it for them, and volunteers will have an easier time organizing events when they are all using the same application. When local governments, non-profits, and volunteers are all connected in one browser application, they can more easily work together to clean up litter.

TrashTag aims to help the environment and the public by providing a resource that clearly highlights the location and type of waste as well as providing live updates regarding existing cleanup efforts and geographic littering patterns to help volunteers allocate their time and resources effectively.

2.2 Major Components (Hardware/Software)

TrashTag uses Python for its backend using Django and JavaScript for its front end using React. Its Application Programming Interface (API) will use Digital Ocean as its host. TrashTag uses MySQL via the Django ORM which uses SQLite.

3 Identification of Case Study

Non-profit organizations and local governments struggle to efficiently locate and remove litter in outdoor recreational areas. Not knowing locations and characteristics of waste items leaves volunteer efforts can be left unprepared and ineffective. Volunteers from non-profit, independent organizations, and individual volunteers struggle to organize cleanups to remove litter efficiently, which wastes time and money. Communities lack a live reporting system to connect citizens with organizations capable of safe and efficient waste removal efforts.

The San Marcos River Foundation is an example of a non-profit organization that would greatly benefit from using TrashTag. They are a local non-profit that struggles to use their volunteers efficiently since they have to find trash along 75 miles of river (The San Marcos River Foundation, n.d.). Knowing exactly where these items were dumped would allow them to use their volunteers more efficiently.

When a user spots litter along the San Marcos River, ideally the user would be able to go on the San Marcos River Foundation Website and click on our link to allow them to use TrashTag. The user would make an account or login, take a photo of the litter, and provide their location or let TrashTag geolocate it for them. Once the location is determined, it will be pinned on a map, and the litter location and photo will be visible to other users. Nearby users will receive a notification that there is nearby litter, and a volunteer cleanup group can be organized or the San

Marcos River Foundation themselves could organize one. Then the team of volunteers will be able to go to the location on the map, take a photo of the cleaned spot or mark it as clean, and the pin will be removed from the map.

TrashTag allows for much more efficient litter cleanup, volunteer organization, and post-cleanup confirmation. This will greatly benefit the San Marcos Rier Foundations' efforts to keep the San Marcos River clean.

4 Glossary

API: Application Programming Interface. A software mechanism for accepting and providing data from and to external applications

Illegal Dumping: The unauthorized disposal of trash and other hazardous materials onto private and/or public property.

Metadata: Descriptive information within data files such as date, location, and timestamp data in photos to be submitted to TrashTag.

Geotagging: The addition of GPS coordinates to images that enable precise location identification where trash was reported.

Live Real-Time Updates: Live data synchronization displaying trash reports to both users and cleanup organizations instantaneously.

Gamification: The incorporation of game-like mechanics such as rankings to motivate user participation and enhance cleanup operations.

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